

MTF Confluence Bot -- Cloud Migration Audit Report

Generated: 2026-04-27 | Server: 129.153.208.32 | OCI Phoenix AD-2 VM.Standard.E2.1.Micro

1. EXECUTIVE SUMMARY -- Cloud Migration Status (2026-04-27)

On April 27, 2026, the MTF Confluence Bot was successfully migrated from a Mac-local process to an Oracle Cloud Infrastructure (OCI) Always Free compute instance. The bot is now fully autonomous -- it does not require the Mac to be on, connected, or awake. All trading decisions, position management, and scheduled analysis now run on the cloud server.

Cloud Server	OCI Phoenix AD-2 129.153.208.32 VM.Standard.E2.1.Micro
OS / Python	Ubuntu 22.04 LTS Python 3.10.12 venv isolated
Bot Status	main.py RUNNING (systemd, autostart) PID active
Writer Status	live_data_writer.py RUNNING (systemd, autostart)
Dashboard Access	http://129.153.208.32:8080/dashboard.html (public HTTP)
Mac Bot Processes	KILLED -- all 5 Mac PIDs terminated 2026-04-27
Mac Crontab	DISABLED -- all jobs migrated to OCI crontab
Alpaca API	Paper account connected paper=True hardcoded
RAM Usage at Start	92 MB / 1 GB limit Headroom: ~900 MB
A1.Flex (12 GB)	NOT provisioned -- Phoenix capacity unavailable. Stack saved, NOT auto-retrying.

2. ARCHITECTURE AUDIT -- Local vs Cloud Component Map

2A. CLOUD (OCI 129.153.208.32) -- Running 24/7 independent of Mac:

Process / Script	Location Schedule Output
main.py --profile paper	CLOUD systemd mtf-bot.service autostart + auto-restart
live_data_writer.py	CLOUD systemd mtf-writer.service autostart + auto-restart
dashboard.html generation	CLOUD generated by main.py every 5-min cycle
scan_results.html generation	CLOUD generated by main.py every 5-min cycle
weekly_review.html generation	CLOUD spawned by main.py post-RTH (1:05 PM PT)
Alpaca order execution	CLOUD broker.py -> paper-api.alpaca.markets
Slack alerts	CLOUD alerts.py -> webhook in .env on OCI
run_market_top.py (cron)	CLOUD cron 5:30 AM PT Mon-Fri output: logs/market_top_latest.json
run_macro_regime.py (cron)	CLOUD cron 5 PM PT Sunday output: logs/macro_regime_latest.json
midday_audit.py (cron)	CLOUD cron 1:15 PM PT Mon-Fri Gemini review -> Slack
nightly_audit.py (cron)	CLOUD cron 1:30 PM PT Mon-Fri Gemini code review -> Slack
options_scanner.py (cron)	CLOUD cron every 15 min 9:30-4 PM PT generates options.html
run_movers.py (cron)	CLOUD cron 5:35 AM PT Mon-Fri pre-market mover scan -> HTML
HTTP server port 8080	CLOUD systemd mtf-http.service serves all HTML dashboards

2B. LOCAL (Mac) -- Can stay local, does NOT affect live trading:

Process / Script	Location Notes
Claude Code / Claude sessions	LOCAL AI dev sessions, code review, board audits
backtest_12pt.py	LOCAL RTH-blocked, runs AH/weekend, no cloud dependency
run_backtest.sh	LOCAL cron disabled, run manually on Mac when needed
Master Brain (NotebookLM)	GOOGLE CLOUD separate from OCI, no bot dependency
preflight_simulation.py	LOCAL dev/testing only
HTML file viewing (browser)	ANY DEVICE http://129.153.208.32:8080/ from any browser

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3. BOARD AUDIT -- CRITICAL FINDINGS & RISKS

[CRITICAL-1] market_top_latest.json MISSING on OCI at launch

main.py loads market_top_latest.json at startup for position sizing (Layer 5). File did not transfer with rsync (empty logs/market_top/ dir on Mac too). Bot defaulted to zone_tier=0 (Green) -- most permissive sizing. run_market_top.py has been triggered manually on OCI to fix. Will run via cron 5:30 AM PT daily going forward.

[CRITICAL-2] macro_regime_latest.json MISSING on OCI at launch

main.py loads macro_regime_latest.json at startup for MIN_SCORE floor (Layer 6). File missing. Bot defaulted to regime_tier=0 (Neutral) -- base MIN_SCORE=9 not adjusted. run_macro_regime.py scheduled for Sunday 5 PM PT cron. Current default is acceptable but stale.

[WARN-1] 1 GB RAM hard ceiling on VM.Standard.E2.1.Micro

Bot started at 92 MB RSS. Normal RTH operation with 3-5 open positions stays under 250 MB. However: weekly_review.py spawned as subprocess adds ~200 MB. Gemini audit scripts add ~150 MB. Simultaneous execution of main.py + options_scanner + midday_audit could reach 700+ MB. OOM kill risk during peak load. MITIGATION: cron jobs are scheduled post-RTH (1:15-1:30 PM PT) when main.py is in its low-activity AH cycle. Monitor with: ssh to server and run 'free -h'.

[WARN-2] A1.Flex (ARM 2 OCPU / 12 GB) NOT auto-retrying

Phoenix capacity exhausted across all 3 ADs. Ashburn blocked by Free Tier 1-region limit. Terraform stack saved (OCID: ocid1.ormstack.oc1.phx.amaaaaaai3ebloyaw73yf3ibme5ldugobra53vwtklhrtd2zy65rl5eoeq) but requires MANUAL Apply clicks in OCI console. Capacity typically opens at off-peak hours (2-5 AM PT). User must manually click Apply -> retry until provisioned. No auto-retry implemented.

[WARN-3] No dead-man's switch / health monitor

If both mtf-bot.service and mtf-writer.service crash and systemd fails to restart them, there is no external alerting. Bot could be down for hours without notification. RECOMMENDATION: Add a cron watchdog that checks service status every 5 min and posts to Slack if down.

[WARN-4] Dashboard on public HTTP (port 8080, no auth)

dashboard.html, weekly_review.html, scan_results.html, options.html are served publicly on http://129.153.208.32:8080/. This is paper account data. No passwords, no HTTPS. Acceptable for paper trading. Before live migration: add nginx + HTTP basic auth.

[INFO-1] PANW / COIN / TOST orphaned shorts adopted at cloud launch

3 positions were open in Alpaca paper account when Mac bot was killed. Cloud bot adopted them correctly at 02:31 UTC with ATR-based stops. PANW short 3sh stop \$180.25, COIN short 3sh stop \$209.69, TOST short 21sh stop \$29.68. All GTC stops are live.

4. PROACTIVE RISK FORECAST -- Tue 2026-04-28 (Tomorrow)

Time (PT)	Risk Level	Issue / Action Required
5:30 AM PT	MEDIUM	run_market_top.py first cloud run -- verify logs/market_top_latest.json exists before 9:30 AM. If it fails, bot enters RTH with zone_tier=0 (Green/permissive). Check: ssh server 'cat logs/market_top_cron.log tail -20'
5:35 AM PT	LOW	run_movers.py first cloud run -- pre-market mover HTML. If FMP API calls hit 250/day limit, movers scan degrades gracefully. Check: ssh server 'cat logs/movers_cron.log tail -10'
9:30 AM PT	HIGH	RTH OPEN with PANW/COIN/TOST overnight shorts. All 3 have GTC stops but market open gap risk is real. PANW earnings on 5/20, COIN earnings on 5/7 -- thesis could shift. Bot will score each position at open

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and may exit if score drops below threshold.

All Day	MEDIUM	GDP Q1 2026 Advance release -- flagged in calendar as HIGH_RISK (50% size event). Bot will auto-throttle sizing. Watch for VIX spike which triggers ATR stop widening.
1:15 PM PT	LOW	First cloud midday_audit.py run. Gemini API key in .env on OCI. Verify Slack receives audit post. If silent: check logs/midday_audit_cron.log
All Day	MEDIUM	RAM headroom watch. If options_scanner.py overlaps with main.py heavy cycle, monitor with: ssh server 'while true; do free -h; sleep 30; done'

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5. BOARD RECOMMENDATIONS -- Prioritized Action List

P0 -- IMMEDIATE

- * Verify market_top_latest.json created on OCI before tomorrow open (5:30 AM PT)
- * Monitor PANW/COIN/TOST short positions at open -- GTC stops in place but gap risk real
- * Test dashboard access: open <http://129.153.208.32:8080/dashboard.html> in browser NOW
- * Confirm Slack is receiving alerts from OCI (not Mac) -- check channel for next bot message

P1 -- THIS WEEK

- * Add systemd watchdog cron: every 5 min check mtf-bot + mtf-writer status, Slack alert if down
- * Manually retry A1.Flex Terraform stack (try 2-4 AM PT when capacity opens) -- OCI console -> Stacks
- * Add run_macro_regime.py immediate run before Sunday cron (regime data stale)
- * Test all 4 HTML pages accessible: dashboard, weekly_review, scan_results, options
- * Verify Gemini midday audit posts to Slack at 1:15 PM PT Tue

P2 -- BEFORE LIVE TRADING

- * Migrate from VM.Standard.E2.1.Micro (1 GB) to VM.Standard.A1.Flex (2 OCPU / 12 GB)
- * Add nginx + HTTP basic auth before exposing any live account data on port 8080
- * Consider Tailscale VPN for private dashboard access from any device without public exposure
- * Set up OCI monitoring alerts (CPU > 90%, RAM > 85%) -> Slack
- * Implement automated rsync from Mac to OCI on code changes (vs manual deploy)
- * Fix GTC-RACE bug in execution/broker.py (open item from 2026-04-22 session)

6. SERVICE STATUS SNAPSHOT -- 2026-04-27 02:31 UTC

Service / Resource	Status
mtf-bot.service	active (running) main.py --profile paper PID 3997 92 MB RAM
mtf-writer.service	active (running) live_data_writer.py PID 3941 6.9 MB RAM
mtf-http.service	active (running) python3 -m http.server 8080 port open
OCI Security List	Port 8080 TCP ingress 0.0.0.0/0 ADDED 2026-04-27
VCN	vcn-20260427-0829 subnet-20260427-0829 (regional) 10.0.0.0/24
Public IP	129.153.208.32 AD-2 PHX VM.Standard.E2.1.Micro Always Free
SSH Key	~/ssh/mtf_bot_oracle (ed25519) ubuntu@129.153.208.32
OCI Crontab	6 jobs: market_top, macro_regime, midday_audit, nightly_audit, options, movers
Mac Crontab	DISABLED -- all 6 jobs commented out with DISABLED-OCI-MIGRATION tag
Mac Bot PIDs	ALL KILLED -- PIDs 25515, 5030, 61392, 61393, 89122 terminated

7. QUICK REFERENCE -- Commands & URLs

SSH to server:	ssh -i ~/.ssh/mtf_bot_oracle ubuntu@129.153.208.32
View live bot log:	ssh server 'tail -f /home/ubuntu/mtf-bot/logs/mtf_bot_stdout.log'
Check service status:	ssh server 'sudo systemctl status mtf-bot mtf-writer mtf-http'
Restart bot:	ssh server 'sudo systemctl restart mtf-bot'
Dashboard HTML:	http://129.153.208.32:8080/dashboard.html
Weekly Review HTML:	http://129.153.208.32:8080/weekly_review.html
Scan Results HTML:	http://129.153.208.32:8080/scan_results.html
Options Scanner HTML:	http://129.153.208.32:8080/options.html
Check RAM:	ssh server 'free -h && ps aux --sort=-%mem head -8'
A1.Flex Stack (OCI):	cloud.oracle.com -> Resource Manager -> Stacks -> mtf-bot -> Apply
Master Brain:	NotebookLM ID: 0203f312-f285-4f20-8b8d-ca6fde65acf7

DISCLAIMER: This is a PAPER TRADING bot. paper=True is hardcoded in execution/broker.py. No real money is at risk. All Alpaca transactions go to the paper account only. This report is generated for internal AI agent review and developer reference.